

DIGITAL PRODUCT DESIGN

Lecture 5: Pricing

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Outline

- Pricing Strategies
- Pricing Analytics
- Pricing Digital Goods

COMMON PRICING STRATEGIES

Pricing Strategies

- Cost-based pricing – Bad
- Competition-based pricing – Bad
- Customer-based pricing – Bad

- Value based pricing – Good

Cost Based Pricing

- Charge a markup over average cost
- Problems
 - Hard to project costs
 - Costs depend on sales, which depend on prices
 - Backwards looking
 - When sales go down, average costs increase. Good time to raise prices?

Cost Based Pricing – Ignores Value Created



Pet Rock - The Original by Gary Dah

[Visit the Pet Rock Store](#)

4.4 ★★★★★ ✓ 602 ratings

100+ bought in past month

\$29⁹⁹

✓prime One-Day

FREE Returns ✓

[Unlock a \\$50 Amazon Gift Card](#) upon app

- A Pet Rock to call your very own!
- Approximately 2 inches but very playful!
- Personal Pet Rock Travel Crate!
- Made with strong material and easy to install and use

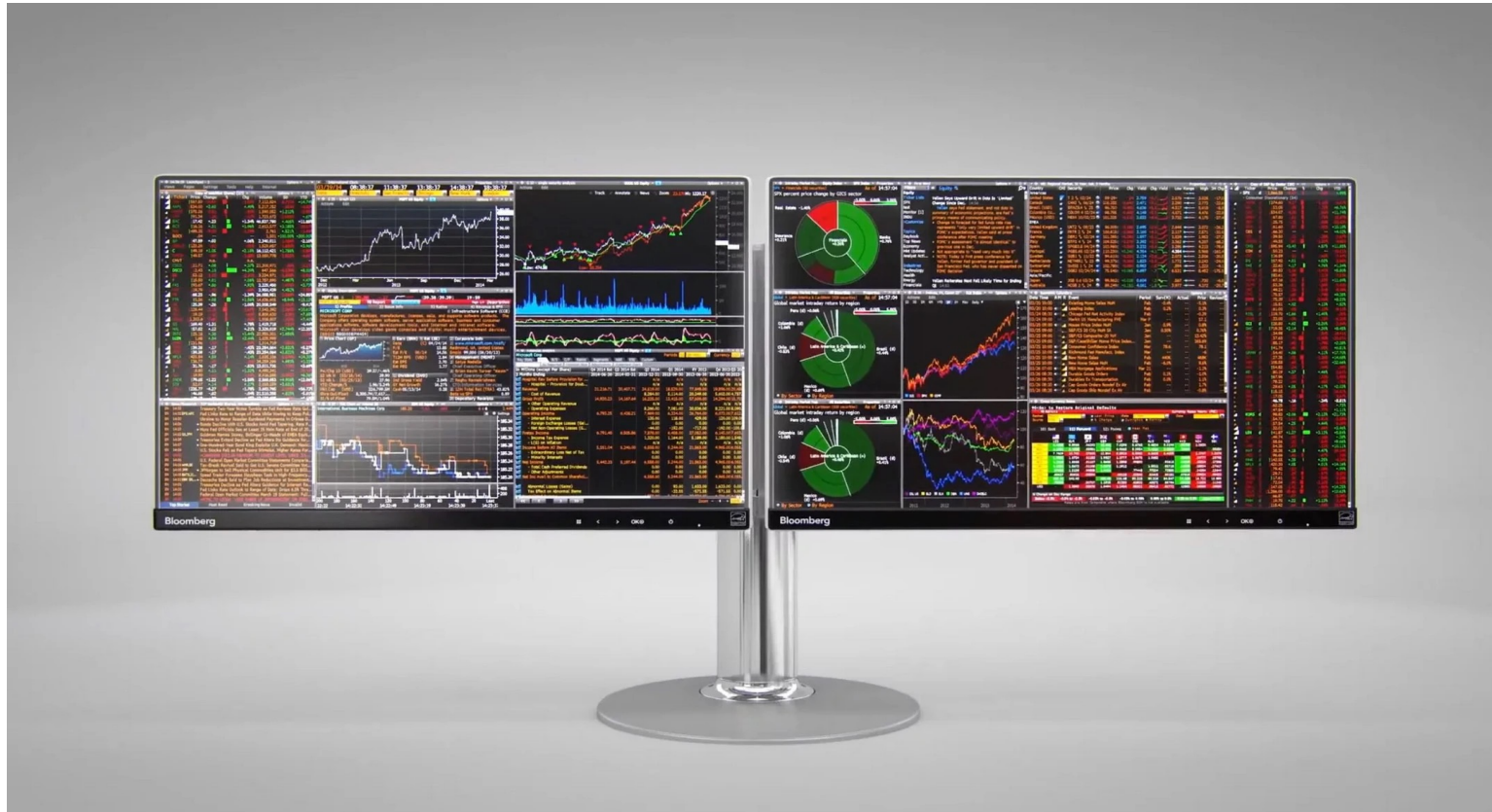
Competition Based Pricing

- Design product, find competitors price, set price at or below competitor
- Problems
 - Ignores unique value proposition
 - Ideally, our product creates more value for our customers!
 - Ignores competitor response
 - If we undercut competitors, will they change their price?

Customer Based Pricing

- Ask customers their valuation and set price based off average
- Problems
 - Customers lie
 - Radiohead “pay what you want” album, 62% paid \$0
 - Average valuation is uninformative
 - We care most about customers willing to pay high prices!

Average Values can be Misleading



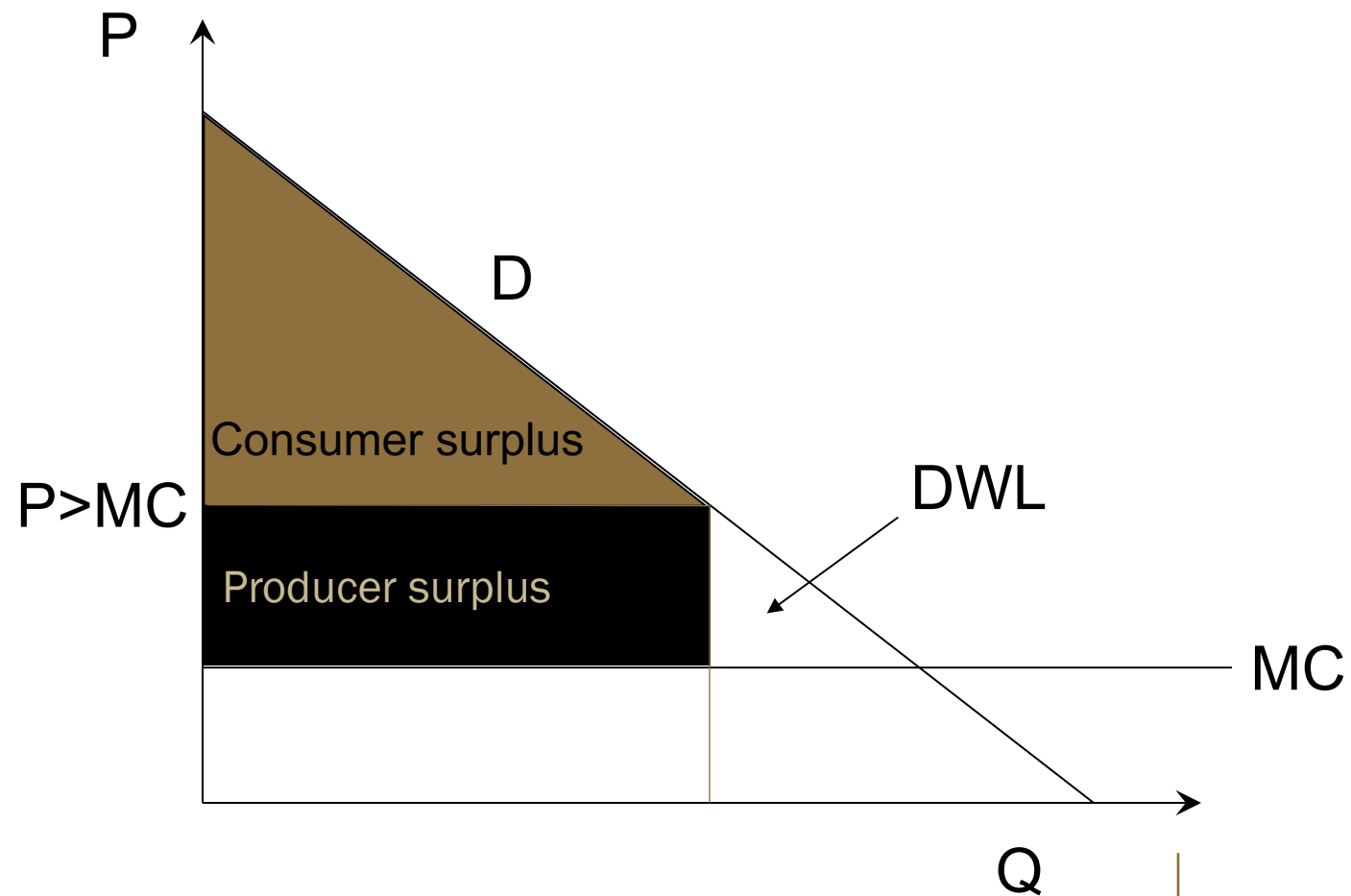
Source: Bloomberg

Value Based Pricing

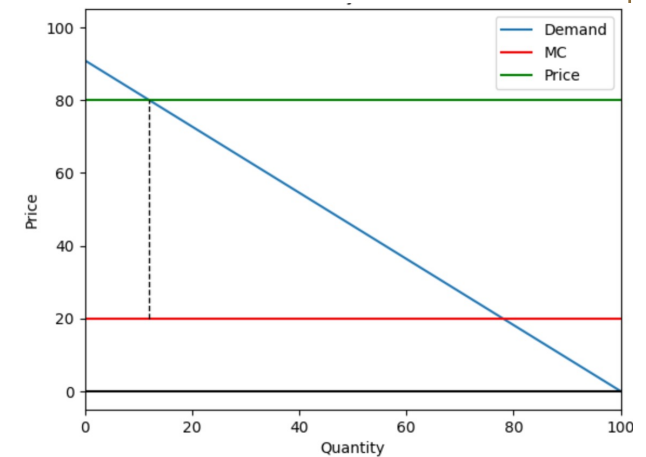
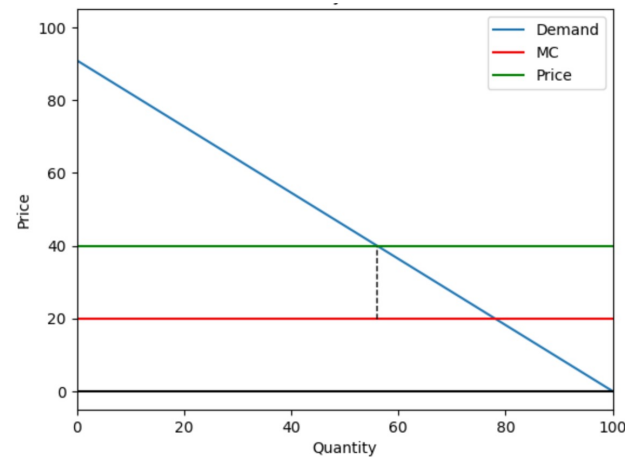
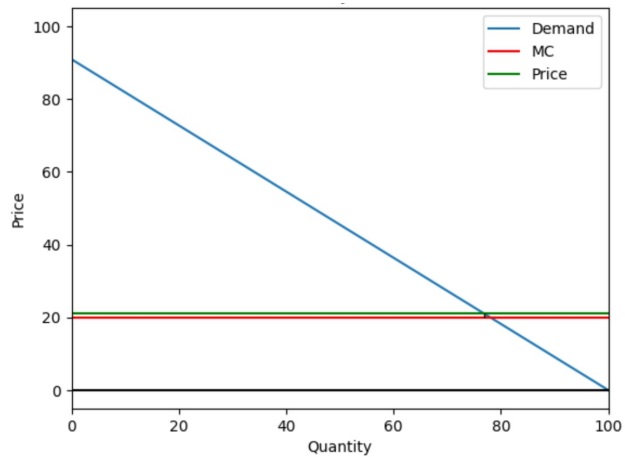
- Estimate a demand curve
 - Demand is the quantity sold at a specific price $Q(P)$
- Economic theory: People purchase if their value for product exceeds the price
 - Demand aggregates individual valuations: $Q(P) = \sum_i 1[V_i > P]$
- Set price to maximize profits:

$$P^* = \arg \max_P P * Q(P) - C(Q(P))$$

Pricing with a Demand Curve



Examples: How Price Impacts Profits



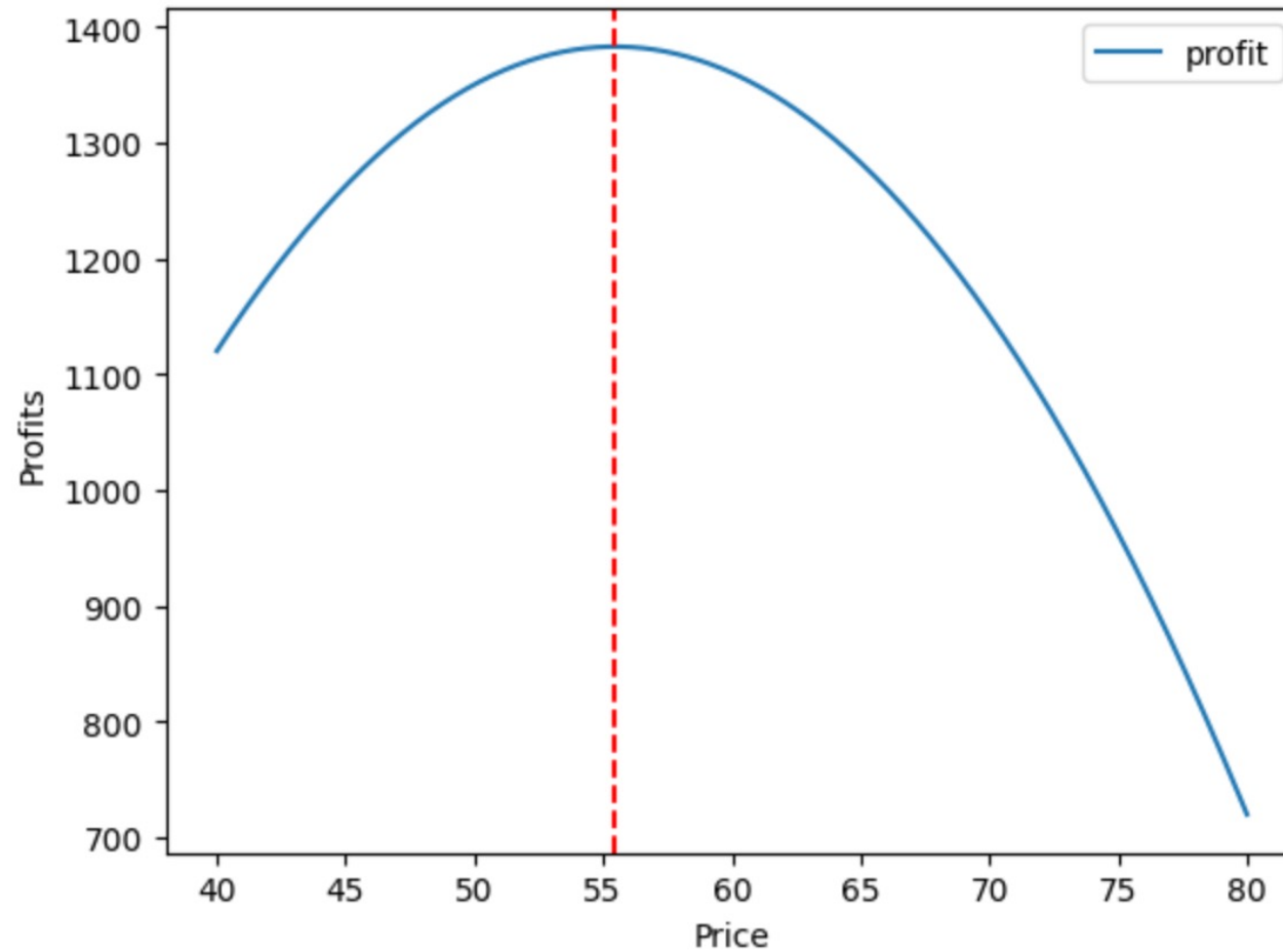
Quick Exercise

- Example on previous slide: $Q(P) = 100 - 1.1 * P, MC = 20$
- Question: Who can find the price closest to optimal?
 - Take 5-10 minutes and calculate price that maximizes profits

$$P^* = \arg \max_P PQ(P) - Q(P) * MC$$

- Good strategy
 - Choose several prices, calculate profits

Quick Exercise, Continued



Price Elasticity is a Key Quantity

- Price elasticity of demand:

$$E_d = \frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Price}} = \frac{\Delta Q / Q}{\Delta P / P}$$

- Under certain conditions:

$$\frac{P^* - MC}{P^*} = -\frac{1}{E_d}$$

- Implications

- If elasticity is close to 0, optimal markups can be large
- If people are price elastic (E_d very negative) optimal markups smaller

Interpreting Elasticities

- $0 > E_d > -1$
 - Great, you have identified a segment of customers who love the product and are price inelastic. Consider raising prices
- $E_d < -3$
 - Common reaction: Great, let's lower prices to increase sales
 - Actuality: Customers price shopping, likely very competitive, reducing prices may cause competitors to respond
- $-1 > E_d > -3$
 - The most common situation. Effect of price change is ambiguous.
 - Likely shouldn't lower prices, as competitors may respond. Possibility to raise.

PRICING IN PRACTICE

Pricing in Practice, Continued

- Given demand curve and costs, we can calculate optimal price
- What do we need to do this?
 - Demand and costs!
- Important that these estimates are causal, need “good” variation in prices

Estimating Demand in Historical Data

- Key questions:
 - Has your firm changed prices?
 - Are these price changes un-correlated with other changes (i.e. demand)?
 - Have you run any pricing experiments?
- If yes, ask your data scientists to estimate demand curve for you!

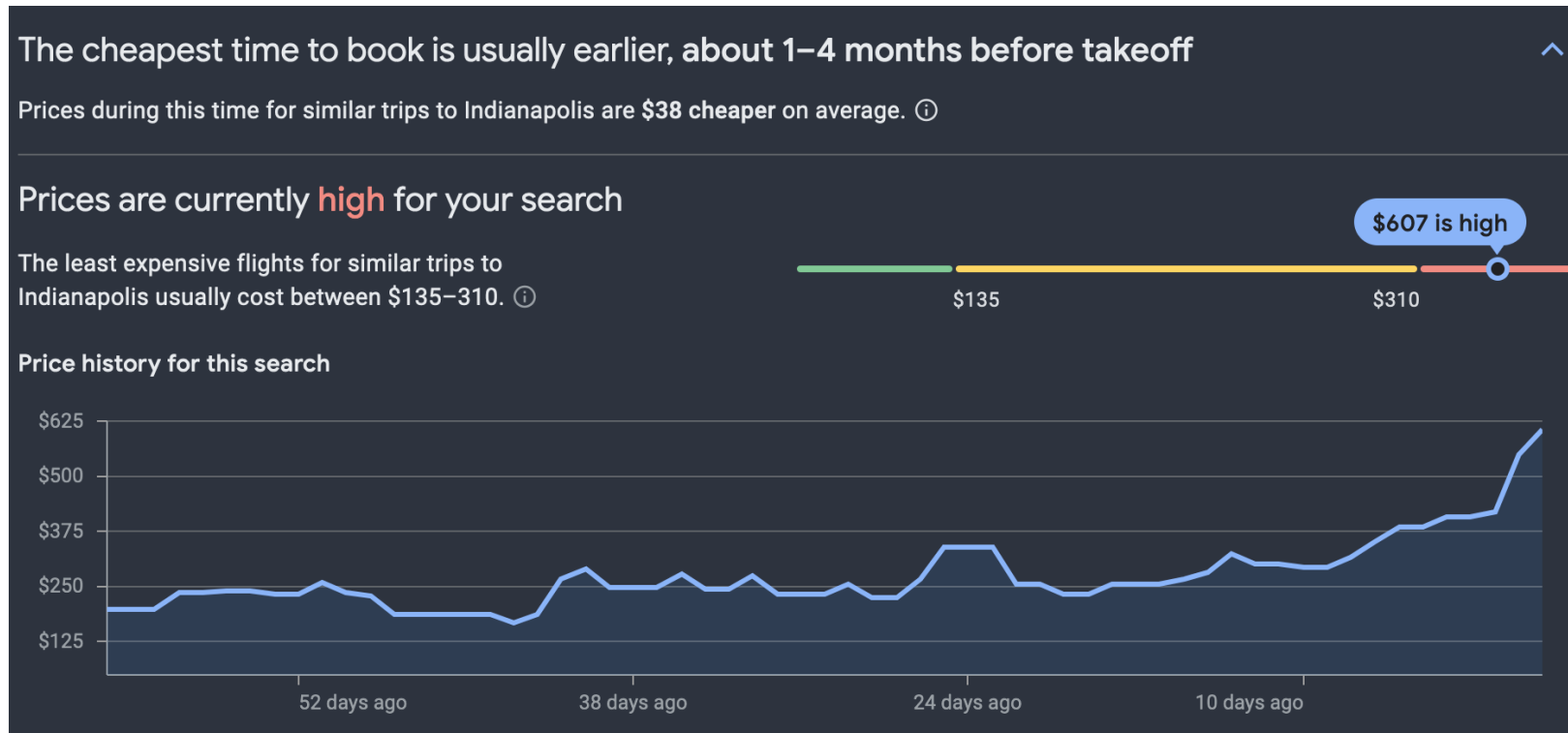
Estimating Demand in New Data

- Are we able to run pricing experiments?
 - If so, create good variation in prices and estimate demand curve
 - Can be risky if consumers see others paying different prices
- Survey experiments next best thing
 - Ask potential customers if they would purchase in a hypothetical environment for a price that is randomly assigned
 - Important to make this as realistic as possible
 - Each person makes a single choice (purchase or not)

Segmenting Customers

- We can also try to predict our high-value customers
 - Customers who value product a bunch
 - Also known as price-inelastic customers
 - If we raise prices, they still purchase
- If we can identify these customers, potential to charge them a different price
 - Known as personalized pricing or “price discrimination”
 - Common methods: Student discount, changing prices over time

Why do Airline Prices Rise Over Time?

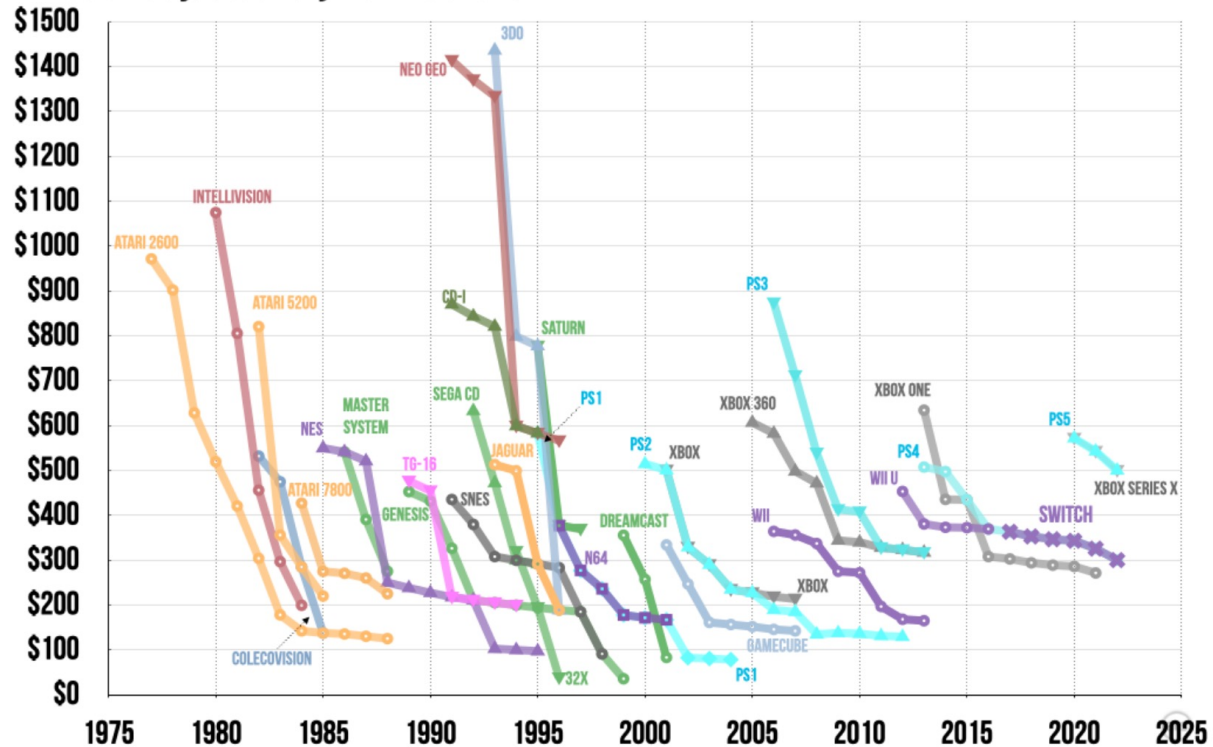


Source: Google Flights

Price Skimming

CONSOLE PRICES OVER TIME

Inflation-adjusted July 2022 dollars



Source: Ken Wilbur

Versioning

LinkedIn Basic:

Connect and network

- **Who's Viewed Your Profile:** Explore 5 most recent viewers in the last 90 days.
- **Insights:** Find and connect with colleagues, classmates, and companies.
- **Search:** Find and view profiles of other LinkedIn members.
- **Profile:** Build a LinkedIn profile to showcase yourself or your company.

Premium Career:

Get hired and get ahead

- **Who's Viewed Your Profile:** Explore all viewers, trends, and insights in the past 365 days. Browse profiles anonymously.
- **InMail:** Message anyone outside your network with 5 InMails per month.
- **Insights:** Get competitive insights about other applicants and companies.
- **Search:** Find the right people with advanced search filters.
- **Profile:** Stand out for more opportunities with AI profile writing assistant.

Try for free

[Learn more](#)

Premium Business:

Grow your network and your business

- **Who's Viewed Your Profile:** Explore all viewers, trends, and insights in the past 365 days. Browse profiles anonymously.
- **InMail:** Message anyone outside your network with 15 InMails per month.
- **Insights:** Get information on companies relevant to your business and industry.
- **Search:** Find the right people with unlimited browsing and advanced search filters.
- **Profile:** Stand out and build credibility with AI profile writing assistant, custom button, and featured profile content.

Try for free

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Versioning, Part 2

BMW / CARS / TRANSP

BMW drops plan to charge a monthly fee for heated seats



Photo by Abigail Bassett for The Verge

/ Customers can now toast their tushies without paying a monthly subscription after the German automaker removed it from its digital store of added features.

By [Andrew J. Hawkins](#), transportation editor with 10+ years of experience who covers EVs, public transportation, and aviation. His work has appeared in The New York Daily News and City & State.

Sep 7, 2023, 2:38 PM EDT

[Link](#) [f](#) [@](#) | [126](#) Comments (126 New)

Source: Ken Wilbur

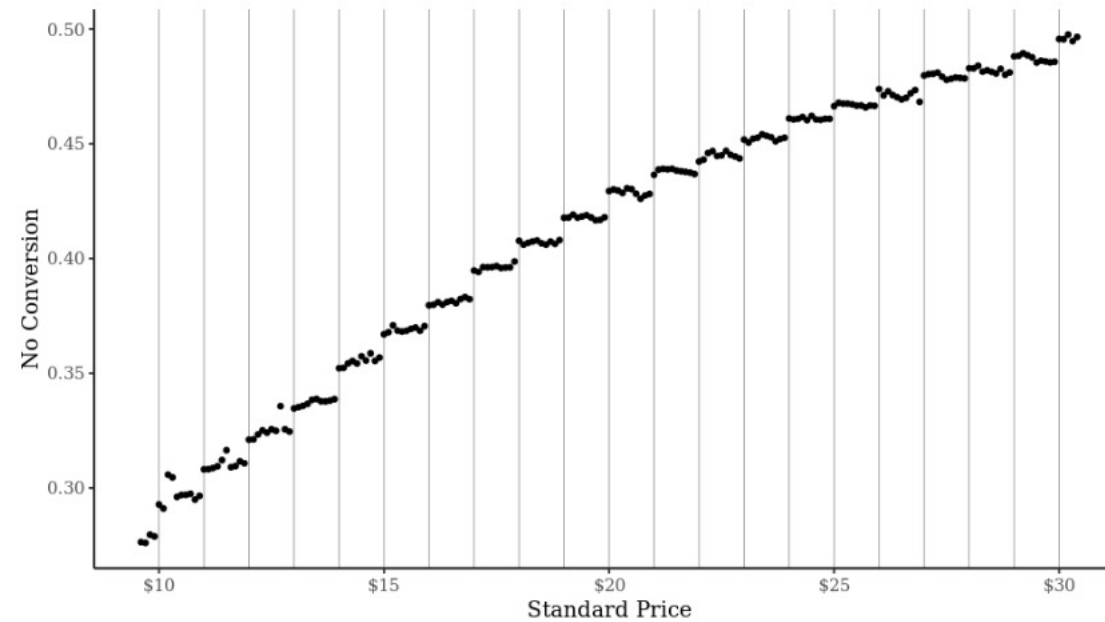
Versioning, Part 3

- Exists in both online and offline worlds
 - E.g. hardcover vs softcover books, first-class vs economy
- Typically have 3 versions for software
 - Take advantage of the compromise effect (next section)
 - If low-end version has $P=0$, this can be a barrier to converting people to paying customers

BEHAVIORAL PRICING

Demand Curves at Lyft – Left Digit Bias

Figure 3: Conversion Rates by Standard Price



Source: List et al (2023)

Compromise Effect



Compromise Effect, Continued



PRICING DIGITAL GOODS

Digital Goods are Unique

- Zero marginal cost
 - Marginal cost of replicating bits is ~ 0
- Non-rival
 - Me watching a movie on Netflix does not preclude you from watching

Implications for Pricing – Bundling

- Bundling is rampant in digital markets, particularly in content
 - E.g. Netflix, YouTube premium, versioning
- Bundling example: Three types of consumers
 - Sports lovers (40%): WTP \$50 for ESPN only
 - Comedy lovers (40%): WTP \$50 for Comedy Central (CC) only
 - Diverse preference (20%): WTP \$30 for ESPN and \$30 for CC

Implications for Pricing – Bundling, Part 2

- Question 1: What is best price to choose if we price ESPN & CC separately?

Example – No Bundling

- How do we set the price of ESPN alone?
 - $P \leq \$30$: Profits = $0.6 * P$, ESPN + Diverse Prefs both purchase
 - $\$30 < P \leq \50 : Profits = $0.4P$, Only ESPN lovers purchase
 - $\$50 < P$: Profits = 0, Nobody purchases
- Best we can do is $P = 50 \Rightarrow \text{Profits} = \20
- Same for CC alone

Implications for Pricing – Bundling, Part 3

- Question 2: What is best price to choose if we price ESPN & CC as a bundle?
 - Meaning we offer 1 price for access to both channels
 - Hint: Diverse preference types value bundle at \$60

Example – Bundling

- How do we set the price of a CC/ESPN bundle?
 - $P \leq \$50$: Profits = P , Everyone purchases
 - $\$50 < P \leq \60 : Profits = $0.2P$, Only diverse prefs purchase
 - $\$60 < P$: Profits = 0, Nobody purchases
- Best we can do is $P = 50 \Rightarrow$ Profits = 50
- Bundling substantially increases profits over pricing each good separately!

Question: Why Bundling So Effective in Digital Goods?

Example – Mixed Bundling

- Mixed bundling occurs when we sell both the bundle and the individual components
- Can further increase profits in this situation
 - $P_B = \$60$, Diverse preferences purchase
 - $P_{ESPN} = \$50$, ESPN lovers purchase
 - $P_{CC} = \$50$, CC lovers purchase
- Capture all the surplus
- Tons of academic research studying different bundling strategies

Increased Price Transparency?

- Many thought that internet would remove cost of comparing prices
- Idea is this would increase competition, lowering prices and prices dispersion
- Didn't necessarily pan out as predicted
 - Price dispersion remains online
 - As search technology improves, so does ability to obfuscate prices
 - Add-on pricing, additional fees, etc

Last Comment on Pricing

- Of all the topics covered, this is the most likely to put you in jail
- Price fixing & bid rigging (auctions) is illegal
 - This includes tacit collusion
- Many examples of extremely wealthy people (with good lawyers) sentenced to jail time for such actions
- Algorithmic pricing does not absolve you from risk

Price fixing by algorithm is still price fixing

By: Hannah Garden-Monheit and Ken Merber

March 1, 2024



Landlords and property managers can't collude on rental pricing. Using new technology to do it doesn't change that antitrust fundamental. Regardless of the industry you're in, if your business uses an algorithm to determine prices, a brief filed by the FTC and the Department of Justice offers a helpful guideline for antitrust compliance: your algorithm can't do anything that would be illegal if done by a real person.

QUESTIONS